

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project presents an end-to-end sales forecasting and demand intelligence system developed using four years of historical retail sales data. Three forecasting models—SARIMA, Facebook Prophet, and XGBoost—were evaluated. XGBoost achieved the best predictive performance and was selected for production deployment. The solution also includes anomaly detection, product demand segmentation, and an interactive Streamlit dashboard to support inventory planning and business decision-making.

Key Findings

Monthly sales showed an overall upward trend with clear yearly seasonality. Technology generated the highest revenue, while the West and East regions consistently contributed strong sales. Time-series decomposition confirmed trend and seasonal patterns, and stationarity testing guided appropriate preprocessing before forecasting.

Forecasting Results

XGBoost outperformed SARIMA and Prophet with the lowest MAE, RMSE, and MAPE, making it the recommended production model. Forecasts indicate continued growth over the next three months, particularly for Technology products and high-performing regions.

Anomaly Detection

Isolation Forest and Rolling Z-Score methods detected unusual sales spikes and drops. Many anomalies likely correspond to festive seasons, promotional campaigns, or exceptional market events. Using two independent techniques improved confidence in anomaly identification.

Product Demand Segmentation

K-Means clustering grouped product sub-categories into High Volume & Stable Demand, Growing Demand, Low Volume & High Volatility, and Declining Demand. These segments enable targeted inventory strategies and improved supply chain planning.

Business Recommendations

1. Deploy XGBoost as the primary forecasting model.
2. Increase inventory for Technology products and high-demand regions before seasonal peaks.
3. Monitor anomalies continuously to respond quickly to unexpected demand changes.

Risk and Limitation

The forecasting models rely on historical sales data and do not account for external factors such as pricing changes, economic conditions, competitor actions, or weather. Regular retraining and the

inclusion of external variables can further improve forecasting accuracy.

Conclusion

The developed system integrates forecasting, anomaly detection, demand segmentation, and interactive visualization into a single business intelligence solution. It supports better inventory planning, demand forecasting, and strategic decision-making while demonstrating practical applications of machine learning and time-series analytics.